

Miscellaneous Topics

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Miscellaneous Topics

- SIADH
- Fever in children
- Baby formulas
- Methanol toxicity
- Q & A
- Mastitis
- Nipple yeast infection

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SIADH

SIADH: Excessive secretion of ADH without a stimuli (such as increased osmolarity or hypernatremia) → water retention and dilutional hyponatremia

Drug-induced SIADH

- Antidepressants: TCAD, SSRI (especially fluoxetine)
- Vinca: vincristine, vinblastine
- AED: carbamazepine and oxcarbazepine
- Others: ACEI, MAOI, phenothiazines, haloperidol, chlorpropamide, nicotine, cyclophosphamide (and ifosfamide) NSAIDs, thiazides, desmopressin, oxytocin, omeprazole, clofibrate

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SIADH

Management of SIADH

- D/C offending therapy (1st line agent)
- Management of hyponatremia:
 - Restrict water to 500 – 1000 mL/day
 - Furosemide to remove water + 0.9% sodium chloride
- Severe hyponatremia: 3% NaCl (hypertonic saline)
- If cannot stop the offending agent:
 - Monitor sodium level
 - Monitor symptoms (Nausea and vomiting, fatigue, restlessness, headache, confusion, loss of energy, muscle spasms and weakness, seizures, loss of consciousness, and coma)
 - Demeclocycline (↑ blood sodium and decrease urine osmolality by inducing nephrogenic diabetes insipidus), oral urea (↑ water excretion in urine), and lithium

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Fever in Children

Temperature

Age	Route
Newborn (or < 5 years with mouth breathing)	Rectal (not for premature babies, ↓ immunity, or patients with rectal problems). Rectal temperature is higher than the mouth temperature, deduct 0.5
Children > 5 years and adults	Oral (not for mouth breathers, and patients with dementia). No hot/cold drinks x 10 minutes
3 months to 5 years who cannot use rectal	Armpit (may add 0.5, inaccurate readings)
Ear: not for children < 2 years. Some inaccuracy converting core temperature to body temperature. ↓ disease transmission, fast, less effect of external factors	

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Baby formulas

Cow milk-based	Common, more protein than breast milk. Whey-based (soluble protein) ↓ emesis compared to casein (curd). Used for full term
Lactose free	Cow-milk based. ↓ mineral absorption and ↑ bacterial colonization of the GIT
Soy protein	lactose free and iron fortified. Used for full term with galactosemia and lactase deficiency and supplemental when mother is vegetarian. Not for infant who develops enteropathy with cow milk
Hydrolyzed protein	Partial (whey formula) – supplemental to breast milk Full (casein formula) – for infant who cannot tolerate cow milk or soy protein, cystic fibrosis, long term diarrhea, cholestasis, short bowel and other conditions that ↓ absorption
Amino Acid based	Contains AA, for severe cow milk intolerants who also cannot tolerate hydrolyzed protein
Thickened formula	Usually useful for infants with excessive regurgitation and also used for weak swallowing reflex (improve swallowing)

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Baby formulas

- Should we use iron-fortified formula?
 - Children ≤ 9 months (for children 6 – 9 months may use iron-fortified cereals instead)
- All children 400 – 1000 IU of vit. D
- Fluid cow milk should be delayed until 9 – 12 months of age (low in iron)

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Baby formulas

- Storage: once opened, at room temp x 1 month in cool dry place
- Mixing: 1 scoop + 60ml of very hot water (tap water boiled x 2 – 5 minutes or water commercially prepared for this purpose). May cool in fridge but must use within 24 hours. Once feeding starts, formula must be used within 2 hours

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Methanol Toxicity

- Methanol level peaks 30-90 minutes following intake but toxicity symptoms appear later
- **Symptoms:** Early: Euphoria, ataxia, headache, nausea, vomiting, epigastric pain followed by drowsiness, ↓ alertness (obtundation), dyspnea, seizures, coma
- **Metabolic acidosis** (which can lead to seizures)
- **Vision loss** (starts as ↓ vision acuity, flashes of light and blurring) due to formic acid accumulation

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Methanol Toxicity

Management of methanol toxicity include all of the following measures EXCEPT:

- A. Fomepizole
- B. IV ethanol
- C. Sodium bicarbonate
- D. Folic acid
- E. None of the above

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Q & A

All of the following drugs and measures can be used for management of diverticular disease EXCEPT:

- A. NSIAD
- B. Low residual diet
- C. Antibiotic
- D. Meperidine

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Q & A

In order to have a similar bioavailability to the brand name, a new generic must have the same ____ as the brand name

- A. AUC
- B. Cmax
- C. Tmax
- D. C_{ss}

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Q & A

A pharmacist works in a poison control center where he receives calls about children who accidentally ingest their parents' medications. The pharmacist would be concerned about which of the following incidences:

- A. A child who swallowed 4 capsules of vancomycin 250mg
- B. A child who ate the whole contents of a nystatin ointment tube
- C. A child who chewed few tablets of simethicone
- D. A child who drank half a bottle of fish oil

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Q & A

What is the course of action of amoxicillin overdose in infants? Monitor for GI symptoms. According to the CPhA monograph "Acute ingestion of large doses of amoxicillin may cause nausea, vomiting, diarrhea and abdominal pain" it also indicates that "For amoxicillin ingestions of less than 250 mg/kg, treatment is not usually required in patients with normal renal function"

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Q & A

Patients allergic to sulfa should be “*careful*” taking all of the following drugs EXCEPT:

- A. Tamsulosin
- B. Sumatriptan
- C. sotalol
- D. Dapsone
- E. Azathioprine

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Q & A

Arylamine Sulfonamides	High cross sensitivity within the class Minimal cross sensitivity with non-arylamine Sulfasalazine, sulfamethoxazole, sulfacetamide, sulfadiazine, amprenavir and fosampranavir
Non-Arylamine Sulfonamides	Minimal cross sensitivity within the class Minimal cross sensitivity with arylamine Carbonic anhydrase inhibitors, SU, diuretics (loop and thiazide), celecoxib, dapson, topiramate, sotalol, sumatriptan, tamsulosin, probenecid

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Q & A

For which of the following infection you may step-down from IV to oral sooner?

- A. Meningitis
- B. Endocarditis
- C. Osteomyelitis
- D. Pyelonephritis

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Q & A

All of the following factors would affect the selection of chemotherapeutic agent in patients' with cancer EXCEPT:

- A. The type of cancer
- B. The stage of the cancer
- C. The patient's age
- D. Comorbidities
- E. Route of drug administration

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Q & A

Which one is more harmful when injected IV?

- A. 5 % dextrose
- B. 3 % dextrose + 0.3 % NaCl
- C. 0.9 % NaCl
- D. Sterile water for injection bag of 250mL
- E. 10 % dextrose

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Mastitis

- Painful inflammation of the breast
- Symptoms: soreness, redness +/- lump. Fever and flu-like symptoms may suggest inflammation and/or infection.
- Causes: Plugged duct due to incomplete draining during breastfeeding (if infected, *S. aureus* is common)

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Mastitis

Risk Factors

- History of mastitis
- Breastfeeding during the first few weeks after giving birth
- Poor breast drainage during feeding due to using one position or switching breasts before it is fully drained.
- Risk ↑ with sore or cracked nipples
- Restriction of milk flow: tight bras, seatbelts, etc.
- Fatigue and stress

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Mastitis

Management

- Increase breastfeeding on the affected side for the 1st 24 hours (may resolve the problem)
- If remains unresolved:
 - Apply warm compress/take warm shower and massage and rest
 - May use analgesic (antibiotic may be needed if infected)
 - Wear supportive bra
 - Drink plenty of fluids
 - CONTINUE breastfeeding (may use the other side if too sore)
- Prevention: Avoid prolonged overfilling of breast, change breastfeeding positions, if breast is not drained, may pump milk.

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Nipple Yeast Infection

- **Symptoms in mother:** Pain (sharp, shooting, radiates to the back or arm described as fire, hot needles, razor blades) during and in-between feeding, sensitive nipple (to touch, clothes, shower, etc.), pink/red, puffy, scaly, weepy and/or blistering nipple. It may also cause mastitis.
- **Symptoms in infant:** white spots or patches with red surrounding on the oral mucous. Infant can get fussy and refrain from breastfeeding. Yeast may overgrow even in absence of symptoms
- **Risk factors:** 1st two weeks postpartum, post-antibiotic use
- Yeast infection of nipple or infant's mouth → ↓ milk release and ↓ nursing efficiency (reversible effect)
- Mother and infant must be treated to avoid transmission even if one is asymptomatic

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Nipple Yeast Infection

Treatment of nipple

- Azoles are 1st choice (↑ resistance to Nystatin) x 2 weeks. Diflucan can be used if topical treatment fails
- Rinse the nipple after breastfeeding and apply the cream 3 – 4 times daily. No need to wipe the nipple before breastfeeding unless can see residues
- **Non drug measures:** Use the least sore side, nurse frequently but for shorter duration, apply ice before feeding can ↓ pain. If nipple is too painful, pump the milk (use within 24 hours), avoid nursing pads (or change daily), rinse the nipple with diluted vinegar.
- **Dietary measures:** yogurt, probiotics (Lactobacillus acidophilus), green tea and ↓ sugar intake

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Nipple Yeast Infection

Treatment of infant

- Nystatin oral suspension x 2 weeks is usually the 1st choice. It should be applied after breastfeeding
- Clean the mouth before applying the medication (drink water or wipe inside with damp washcloth)
- Apply ½ the dose in each cheek (Put the medicine in a spoon and use a Q-tip to swab it in the baby's mouth)
- **Non Drug Measures:** Boil any object that goes in the baby's mouth (such as pump parts, bottles, toys, and pacifiers) for 20 minutes each day. Add some vinegar to the boiling water.

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Nipple Trauma

Nipple trauma

- Fissures, ulceration, erythema, edema, blisters, pain or bruising in the nipple or areola area.
- The main cause is improper latch of the baby to the breast.
- Lanolin before and after each feeding (or olive oil before feeding) can be used to ↓ trauma.
- Silicone nipple shield can be used to cover the sore nipple. It improves latch but ↓ milk flow on the long run.

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